H

TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CIVICS GROUP	1	2				TUTOR NAME	

Biology

9648/02

Paper 2

Monday, 2 September 2013

2 hours

Candidates answer on the Question Paper

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

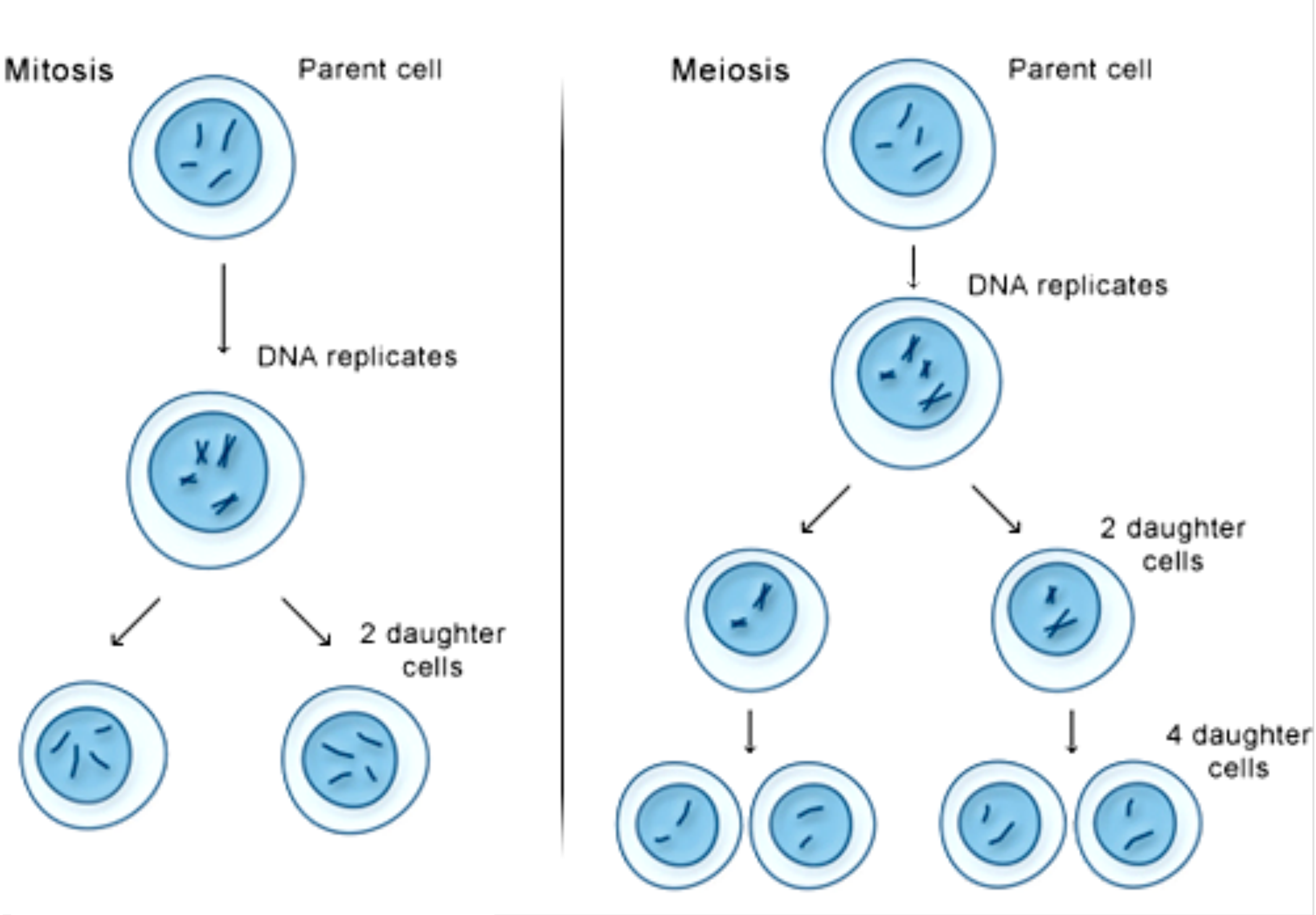
For Examiner's Use		
Section A		
1	10	
2	10	
3	17	
4	11	
5	10	
6	9	
7	13	
Section B		
8 OR	20	
9	20	
Total	100	

SECTION A - STRUCTURED QUESTIONS (80 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION 1

Fig. 1.1 shows a simplistic outline of the processes of mitosis and meiosis.



(a) State four ways in which the behavior of chromosomes in meiosis differs from their behavior in mitosis. [4]

Fig. 1.2 shows the karyotype of a person who suffers from a double genetic disorder due to the inheritance of extra copies of chromosomes.

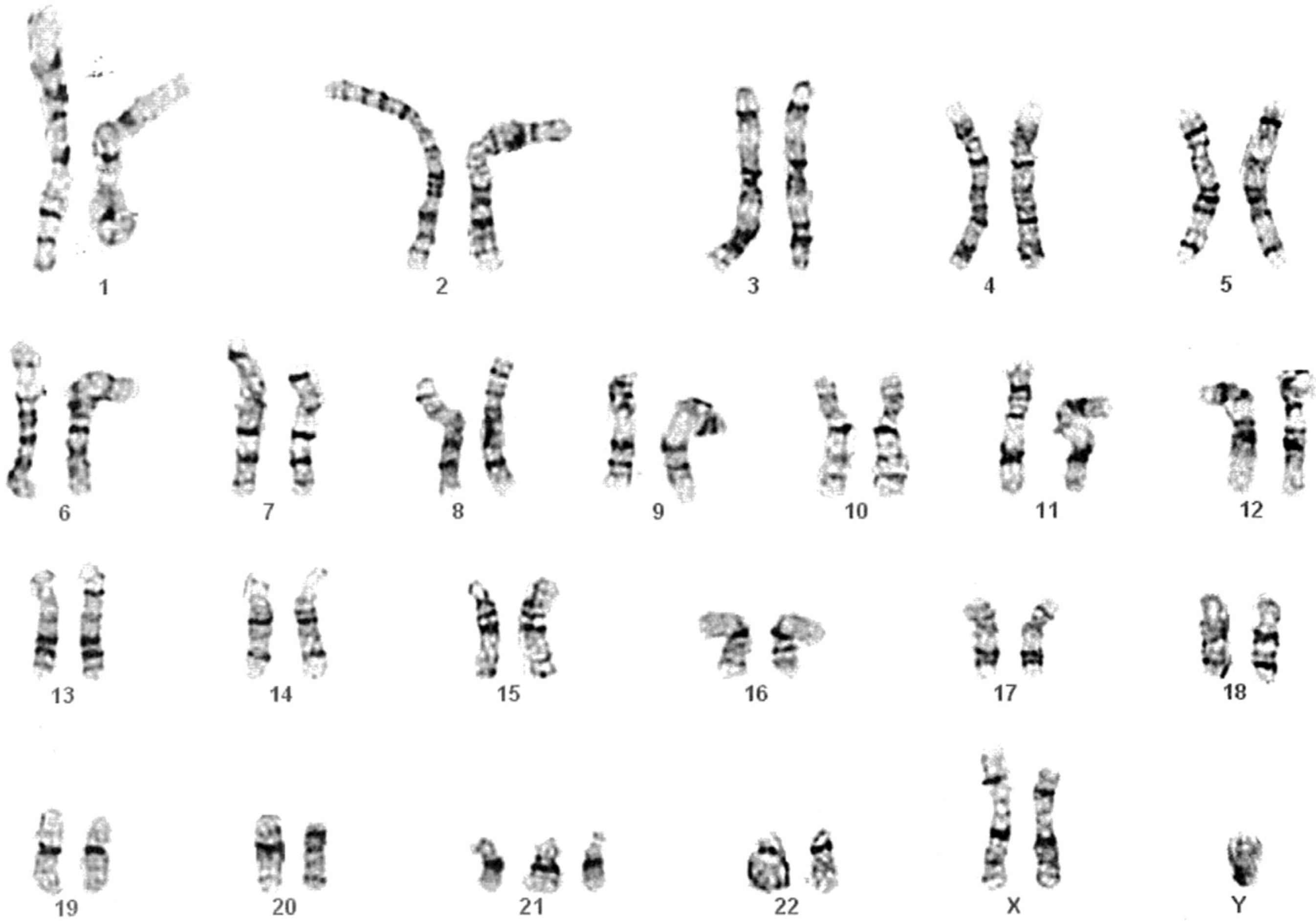


Fig. 1.2

(a) With reference to **Fig. 1.2**, suggest the genetic disorder(s) the person whose karyotype is shown is suffering from. [1]

(b) Describe the aberration that has occurred during cell and nuclear division that has resulted in the inheritance of the extra chromosomes. [3]

(c) Explain the effects of having extra copies of chromosomes on the person. [2]

[Total: 10]

QUESTION 2

Fig. 2.1 is a diagram showing part of a bacterial chromosome with a single origin of replication and two replication forks which proceed in opposite directions until replication of the entire bacterial chromosome is complete.

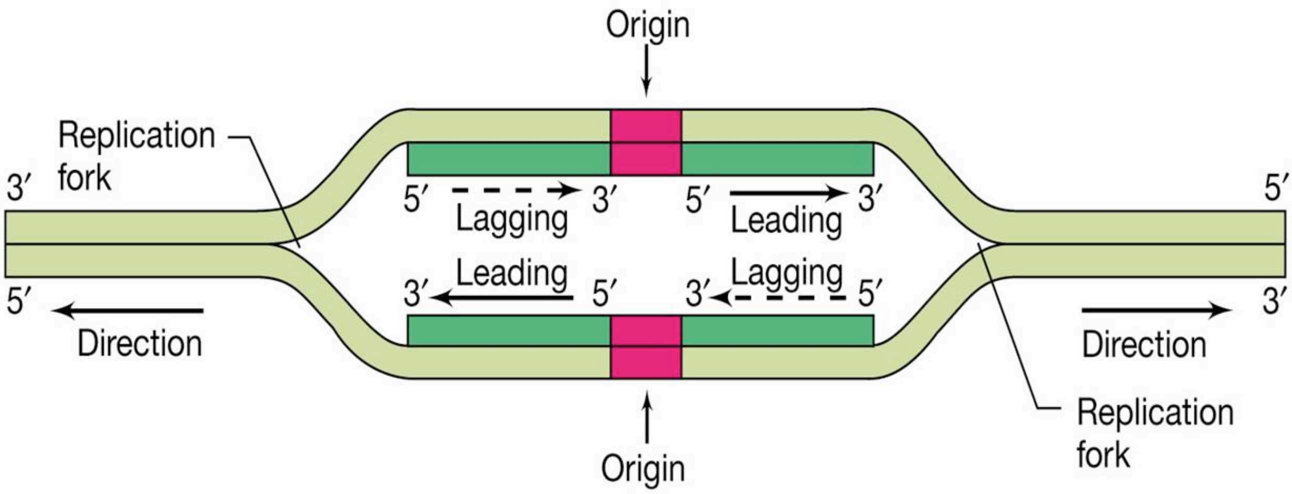


Fig. 2.1

(a) With reference to **Fig. 2.1**, explain why this replication is said to be semi-conservative. [2]

(b) With reference to **Fig. 2.1**, replication only occurs in a 5' to 3' direction, explain why this is so. [2]

The Hayflick limit is the number of times a normal human cell population will divide until cell division stops. Empirical evidence shows that the telomeres associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. The Hayflick limit has been found to correlate with the length of the telomere region at the end of a strand of DNA. **Fig. 2.2** shows the position of telomeres in a DNA double helix.

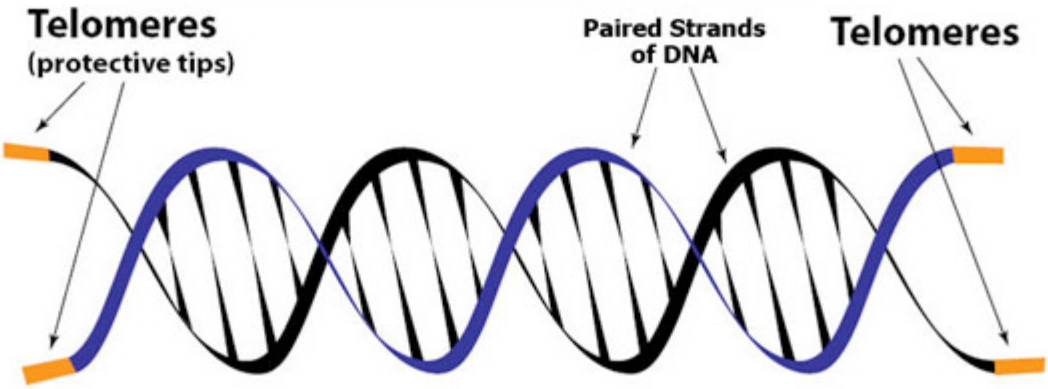


Fig. 2.2

(c) Outline the role of telomeres. [3]

(d) Explain how telomeres shorten with each cell division. [3]

[Total: 10]

QUESTION 3

Fig. 3.1 shows a signaling pathway involving Ras protein, a proto-oncogene. Ras protein is involved in the activation of PI3K a kinase which is a relay molecule for the downstream activation of subsequent molecules within the phosphorylation cascade. The effect of this signaling pathway is the triggering of cell growth, proliferation and survival.

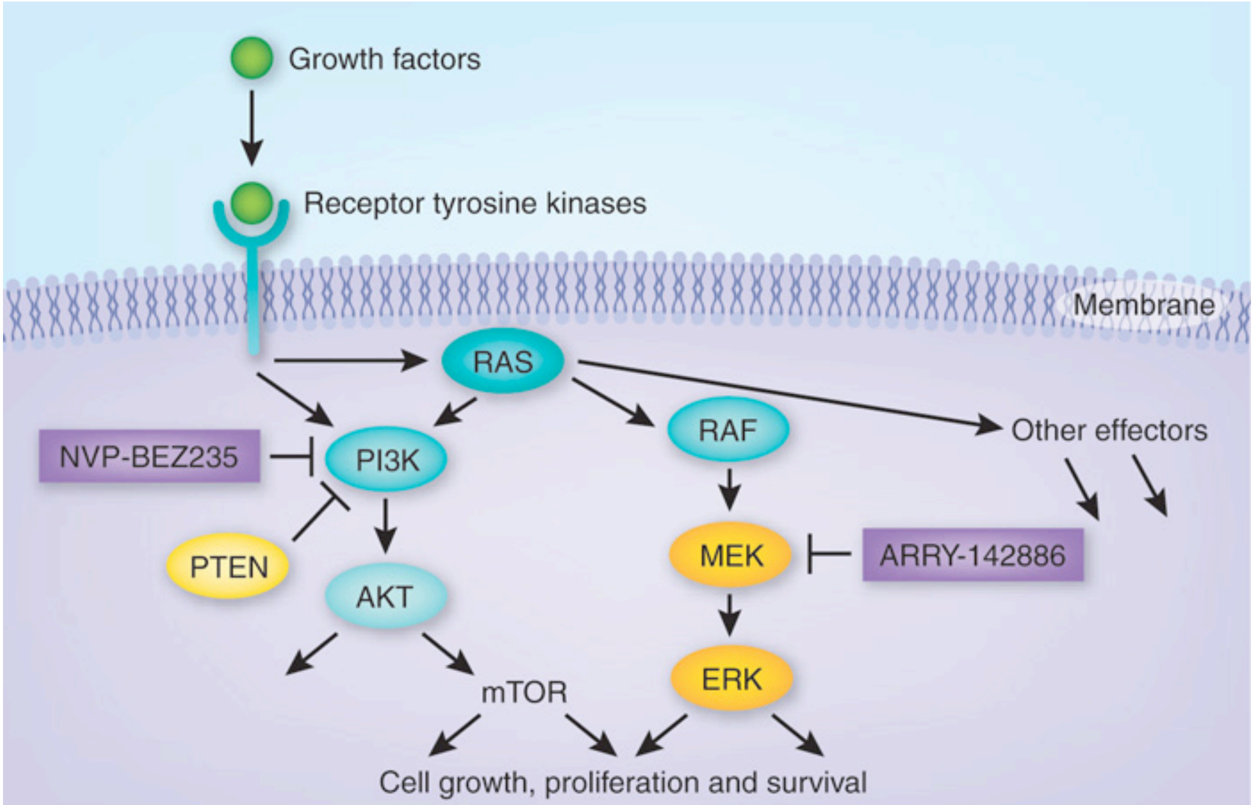


Fig. 3.1

(a) With reference to **Fig 3.1**, describe how binding of Growth Factors could result in the activation of Ras protein. [2]

Ras proto-oncogene could be mutated such that it becomes oncogenic. This mutation of *Ras* proto-oncogene is frequently seen in many cases of cancer. However, in order for oncogenic Ras protein to induce cancer formation, another protein, PTEN, must be mutated. In the signaling pathway shown in **Fig. 3.1**, PTEN is involved in the inhibition of PI3K and thereby signals cells to stop dividing and can cause cells to undergo programmed cell death (apoptosis) when necessary.

(b) State what you understand by the terms,

(i) Proto-oncogene [1]

(ii) Onco-gene [1]

(c) Explain how mutation could make *Ras* oncogenic. [4]

PTEN is an enzyme that acts as a tumour suppressor by regulating cell cycle and induces apoptosis. PTEN inhibits PI3K, a kinase, by dephosphorylating PI3K (refer to **Fig. 3.1)**. Dephosphorylated PI3K is inactive and will not activate downstream relay molecules of the phosphorylation cascade to induce cell division. **Fig. 3.2** shows the structure of PTEN.

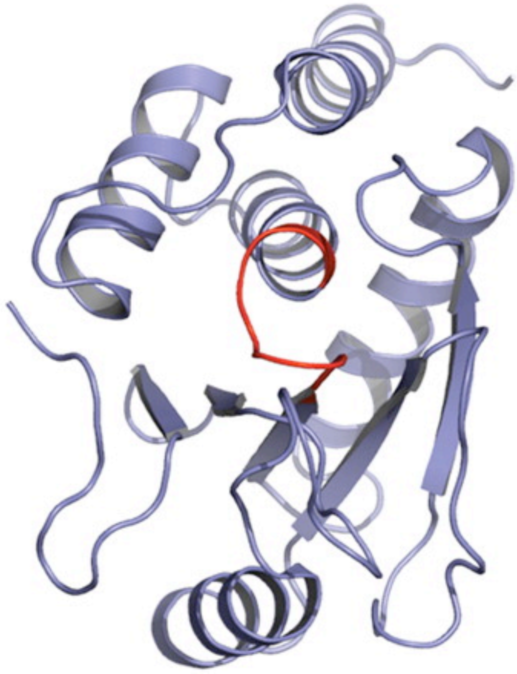


Fig. 3.2

(d) Explain how a mutation in the PTEN gene could affect the structure of its product. [3]

(e) Suggest why some mutations of the PTEN gene have no effect on the activity of the PTEN protein. [3]

Fig. 3.3 shows another protein, P53, which has tumour suppressor activities, binding to DNA. p53 works as a transcription factor by binding to DNA to trigger the transcription of a series of genes which work in suppressing tumour formation.

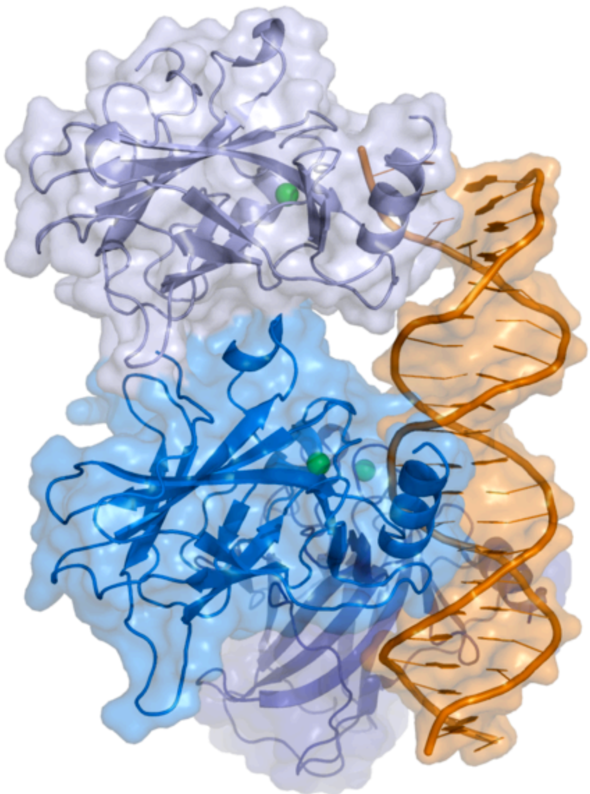


Fig. 3.3

(f) Suggest why the p53 protein only binds to certain parts of the DNA molecule. [3]

[Total: 17]

QUESTION 4

Fig. 4.1 shows some of the main structural features of the H7N9 influenza virus.

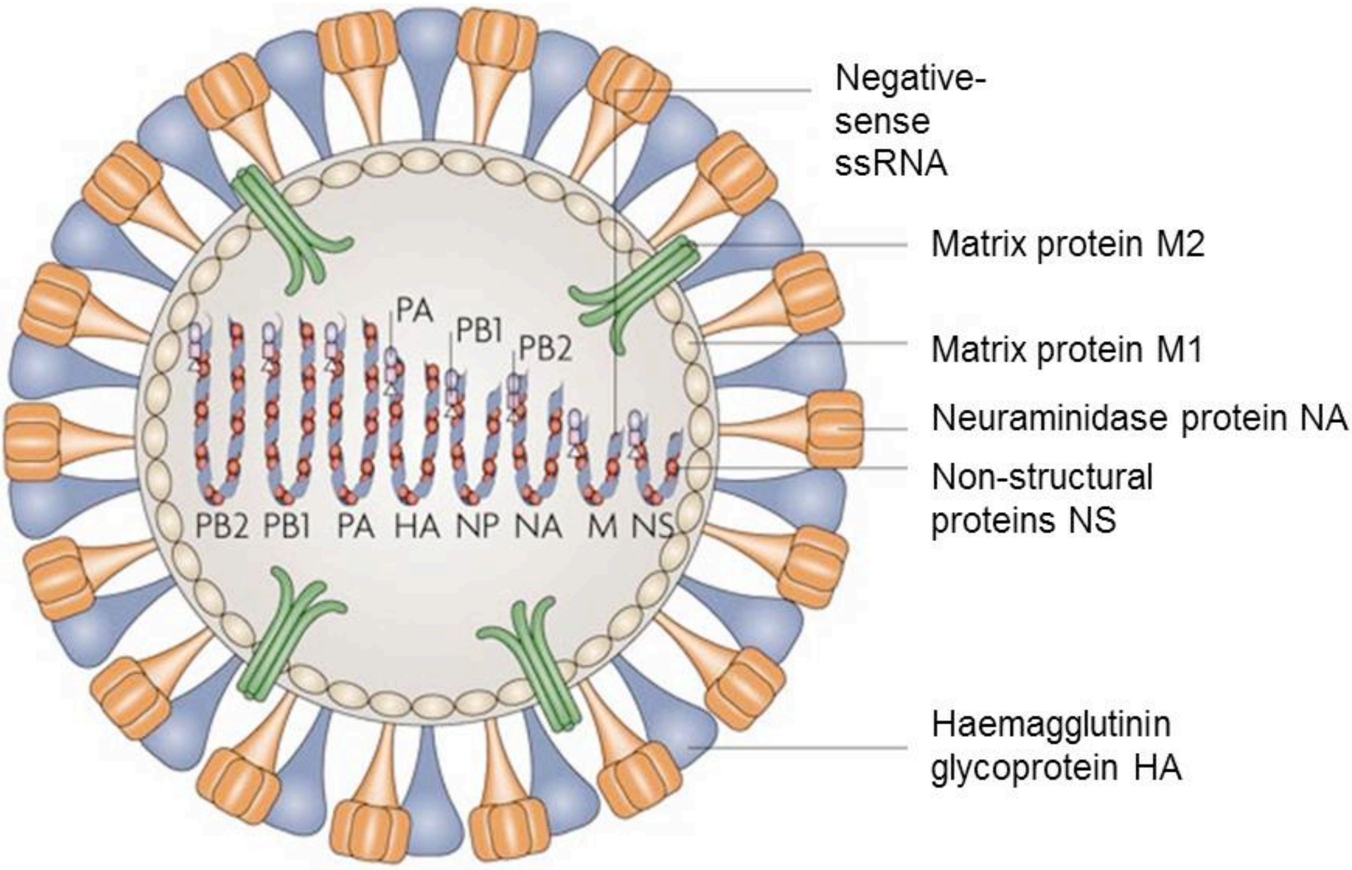


Fig. 4.1

The viral RNA genome contains genes (*HA* and *NA*) that codes for Haemagglutinin and Neuraminidase, which are widely researched for the development of antiviral drugs.

In H7N9, the "H" refers to the type of haemagglutinin and the "N refers to the type of neuraminidase present as part of the viral envelope. The avian influenza A (H7N9) virus designation of H7N9 identifies it as having HA of the H7 subtype and NA of the N9 subtype

After the influenza virus enters the host cells, the genes in the RNA segments are used as templates for mRNA production

(a) Suggest the functions of haemagglutinin and neuraminidase in Influenza virus. [2]

____

____

____

(b) Describe how an enveloped virus such as H7N9 virus enters a host cell. [4]

(c) The viral RNA also contains *PB2, PB1* and *PA* which code for the viral RNA polymerases. Explain why such RNA segments are needed by the virus. [2]

(d) The influenza virus does not contain ribosomes. Despite this, the influenza virus is able to trigger synthesis of viral proteins. Explain how does the influenza virus do so? [1]

H7N9 is an Avian influenza virus which normally circulates amongst avian populations with some variants known to occasionally infect humans. An H7N9 virus was first reported to have infected humans in 2013 in China and now this virus has spread among humans. **Fig. 4.2** show the evolution of H7N9 virus.

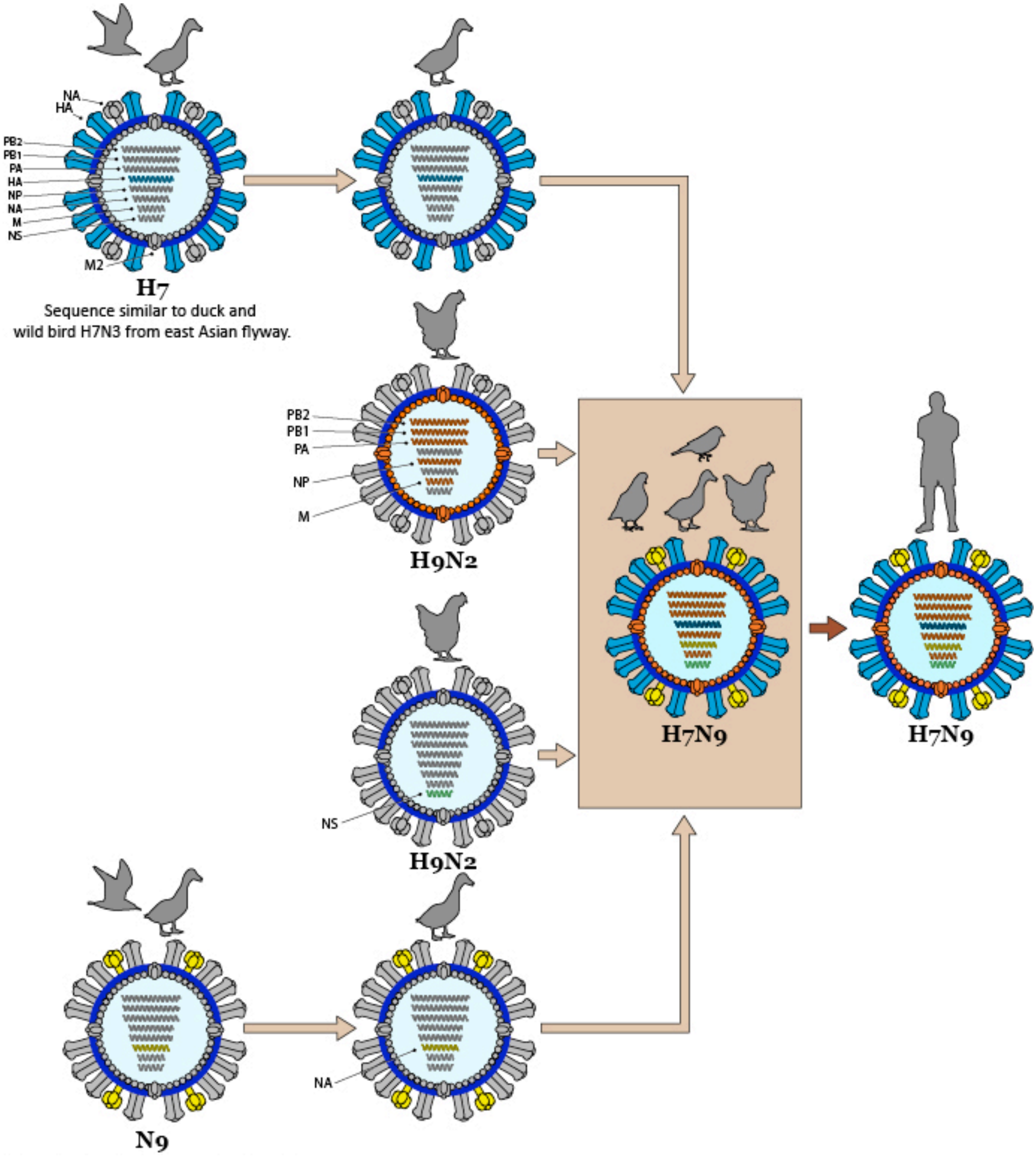


Fig. 4.2

(e) Within reference to **Fig. 4.2**, explain how the new combination of RNA segments in H7N9 could have arisen. [2]

[Total: 11]

QUESTION 5

Carbon is essential to life. All of our molecular machines are built around a central scaffolding of organic carbon. Unfortunately, carbon in the earth and atmosphere is locked in highly oxidized forms, such as carbonate minerals and carbon dioxide gas. In order to be useful, this oxidized carbon must be "fixed" into more organic forms, rich in carbon-carbon bonds and decorated with hydrogen atoms. Powered by the energy of sunlight, plants perform this central task of carbon fixation.

Inside plant cells, the enzyme ribulose bisphosphate carboxylase (rubisco) forms the bridge between life and the lifeless, creating organic carbon from the inorganic carbon dioxide in the air. **Fig. 5.1** shows the structure of ribulose bisphosphate carboxylase (rubisco).

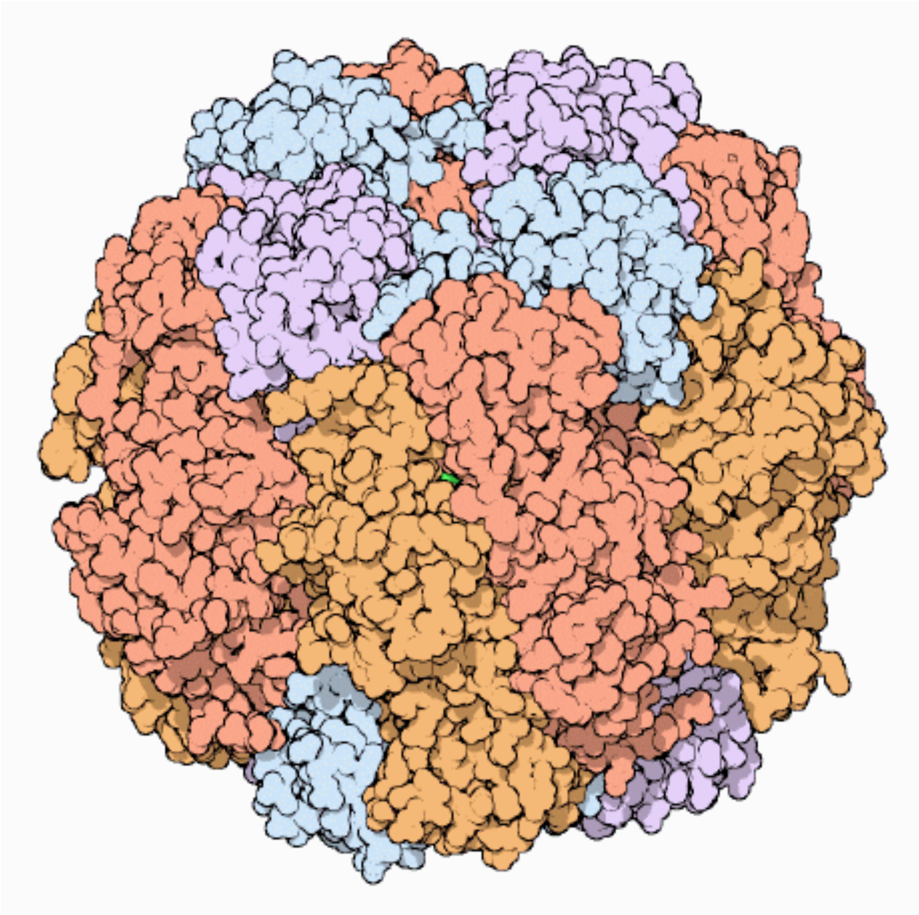


Fig. 5.1

(a) Rubisco is an enzyme made of sixteen polypeptide chains.

Describe how the structure of rubisco is maintained. [2]

In spite of its central role, rubisco is remarkably inefficient. Typical enzymes can process a thousand molecules per second, but rubisco fixes only about three carbon dioxide molecules per second. **Fig. 5.2** shows the relationship between CO_2 assimilation rate and increasing light intensity in a plant, when carbon dioxide is present in excess.

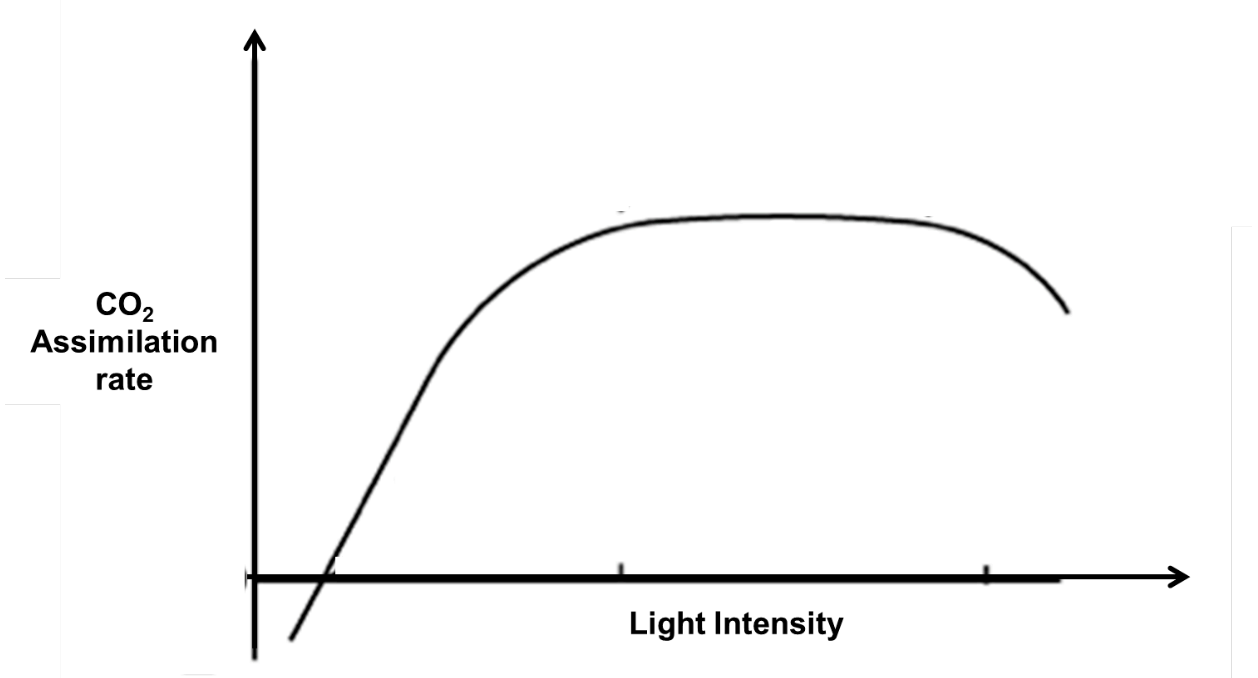


Fig. 5.2

(b) Explain how light energy assists in the process of carbon fixation in plants. [3]

(c) Explain why the CO_2 assimilation rate levels off at higher light intensity. [3]

Rubisco also shows an embarrassing lack of specificity. Unfortunately, oxygen molecules and carbon dioxide molecules are similar in shape and chemical properties. In rubisco, an oxygen molecule can bind comfortably in the active site designed to bind carbon dioxide. Rubisco then attaches the oxygen to the sugar chain, forming a faulty oxygenated product.

(c) With reference to **Fig 5.2**, suggest why the CO_2 assimilation rate is declining as the light intensity increases higher. [2]

[Total: 10]

QUESTION 6

Hair color is the pigmentation of hair follicles due to two types of melanin: eumelanin and pheomelanin. Pheomelanin colors hair reddish-orange. Eumelanin, which has two subtypes of black or brown, determines the darkness of the hair color. Generally, if more eumelanin is present, the color of the hair is darker. If less eumelanin is present, the hair is lighter. Particular hair colors are associated with ethnic groups.

The genetics of hair colors are not yet firmly established. According to one theory, at least two unlinked gene pairs that control human hair color.

In Americans, gene **B** and **R** are involved in controlling the shades of hair colour. Gene **B** controls the production of brown eumelanin. This gene has a dominant brown allele and a recessive blond allele. Gene **R** controls the production of pheomelanin. This gene has a non-red allele (which suppresses production of pheomelanin) which is dominant and the allele for red hair is recessive. Gene **B** is epistatic over Gene **R**.

(a) State and explain the genotypes for a blonde hair phenotype. [3]

A couple, both with blonde hair, was filing for a divorce as the husband felt that her wife has cheated on him. His rationale of saying so was based on the recent birth of their boy who has red-coloured hair.

(b) With the use of a genetic diagram, explain whether or not it is possible for two parents with blonde hair to have an offspring with red-coloured hair. [4]

(c) The husband felt bad for doubting his wife. He wants to find our whether there is a significant difference between what they observed in their baby boy and what is expected.

Explain whether it is possible to carry out a chi-square test in this case. [2]

[Total: 9]

QUESTION 7

Fig. 7.1 shows Honeycreepers of the Family Drepanididae from the Hawaiian Islands. These Honeycreepers reside on the various Hawaiian Islands. Through a series of studies, these Honeycreepers are found to be evolved from an ancestral Honeycreeper which resided originally on, Kauai, one of the Hawaiian Islands.

Fig. 7.1

(a) Explain how these various species of Honeycreepers could have arisen. [4]

Our understanding of the phylogeny of species has been greatly improved through the use of molecular techniques.

Many phylogenetic studies use mitochondrial DNA rather than nuclear DNA.

(b) Explain what is meant by the term phylogeny. [2]

A new Honeycreeper (**Fig. 7.2**) has been found on one of the Hawaiian Islands. Based on its structural features, Scientists are speculating that this new found Honeycreeper is also a descendant of the ancestral Honeycreeper found on Kauai.

Fig. 7.2

(d) Suggest why it is not advisable to determine the phylogenetic relationship of this new found Honeycreeper to the rest of them from the morphological features. [2]

(e) Outline how DNA can be studied to determine phylogeny of organisms. [5]

[Total: 13]

SECTION B (20 MARKS)

Answer **one** question.
Write your answers on the separate writing paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION 8

(a) Compare anaerobic respiration and aerobic respiration. [6]

(b) Describe the main features of oxidative phosphorylation. [8]

(c) Discuss how the mitochondrion is adapted for carrying out respiration. [6]

QUESTION 9

(a) Compare the mode of infection of Influenza Virus and Human Immunodeficiency Virus. [6]

(b) Describe the mechanisms of gene transfer in prokaryotes. [8]

(c) Discuss how the prokaryote, E. coli, is able to respond to varying concentrations of lactose. [6]

End of Paper

BLANK PAGE